

**A DUAL-INPUT BIDIRECTIONAL LSTM AND CAUSAL
LINEAR REGRESSION SYSTEM FOR SOLAR SALT
CRYSTALLIZATION FORECASTING IN PUTTALAM, SRI
LANKA**

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DECLARATION

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ABSTRACT

Sri Lanka's solar salt industry in Puttalam has seen a drop in production of more than 40% from 2023 to 2024 due to unseasonal monsoon rains and climate change getting worse. Saltern operators still use manual logbooks and their own experience to decide when to crystallize and harvest, and there is no data-driven decision support available. This individual report shows Crystal, which is the Crystallization Forecasting part of BrineX, an AI platform made in collaboration with the Puttalam Salt Society (PSS).

Crystal addresses the core predictability problem in solar salt production by deploying a dual-input Bidirectional Long Short-Term Memory (LSTM) network trained on 1,095 daily records of eight pond crystallization parameters and fourteen OpenWeatherMap weather features, forecasting daily crystallization conditions over a 60-day horizon. A causally-specified Linear Regression model then turns LSTM daily outputs into monthly production volume forecasts. It gets a R^2 of 0.9732 on 36 months of PSS historical data, which is a statistically significant improvement of 0.5632 R^2 over the ARIMA baseline. The trained LSTM model is serialised to ONNX format and served through ONNX Runtime in a NestJS microservice, achieving zero failed requests across 3,112 stress test invocations and an average response time of 88.80ms under peak load.

The system provides saltern operators with a daily crystallization readiness dashboard, a traffic-light alert system (Red/Yellow/Green), seasonal production outlooks, and confidence-bounded forecasts. Model selection was conducted through empirical rejection of ARIMA, Holt-Winters, Prophet, Random Forest, and XGBoost alternatives, with the final architecture selected based on superiority in both predictive accuracy and causal interpretability.

Crystal is the first published data-driven system for predicting how Sri Lankan solar salt will crystallize. It directly supports national food security and helps the larger BrineX platform reach its goal of turning reactive, manual saltern operations into proactive, climate-resilient production planning.

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LIST OF ABBREVIATIONS

Table 0.1: List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DaaS	Data-as-a-Service
ERP	Enterprise Resource Planning
GPU	Graphics Processing Unit
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
RMSE	Root Mean Square Error
SaaS	Software-as-a-Service
UI	User Interface
AWS	Amazon Web Services
GCP	Google Cloud Platform
ADB	Asian Development Bank
UNDP	United Nations Development Programme
ML	Machine Learning
MODIS	Moderate Resolution Imaging Spectroradiometer
NOAA	National Oceanic and Atmospheric Administration
ONNX	Open Neural Network Exchange
PSS	Puttalam Salt Society
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
SaaS	Software-as-a-Service
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Variables
SQL	Structured Query Language
SSA	Singular Spectrum Analysis
UI	User Interface

1. INTRODUCTION

1.1. BACKGROUND AND LITERATURE SURVEY

One of the oldest industrial processes still in use today is solar evaporation. It uses the natural combination of high temperatures, low rainfall, and strong winds to gradually concentrate seawater brine until sodium chloride reaches supersaturation and forms solid salt crystals. Historically, this process has been the basis of Sri Lanka's national salt self-sufficiency. Major salterns in Puttalam, Hambantota, and Elephant Pass have served both domestic and regulated export markets [1]. The Puttalam salterns, which are mostly run by the Puttalam Salt Society (PSS), are the biggest and most complicated of these sites. They have thousands of shallow evaporation ponds that are run by individual landowners during two main production seasons.

The two main seasons that affect the production of solar salt at Puttalam are the Maha season (December to March), when the weather is drier and cooler, which is better for high-purity crystallisation, and the Yala season (May to September), when the weather is warmer but the amount of rain varies more. The northeast monsoon usually stops or greatly slows down production during the transition period from October to November. Crystallisation is the most environmentally sensitive step in the whole production process. This is when dissolved sodium chloride reaches supersaturation and falls out of concentrated brine. Temperature, humidity, wind speed, rainfall intensity, and pond salinity all work together to decide if a certain day or week is good for continuing to concentrate brine or if there is a risk of dilution, crystal dissolution, and loss of harvest [2].

In climates that have been stable for a long time, experienced saltern operators could use their knowledge of past seasons to predict when crystallisation would happen and plan their harvests accordingly. But the fact that the weather has become less predictable over the past ten years has made this experience-based method structurally useless. The year 2023–2024 saw 1,895 mm of rain fall in Sri Lanka's saltern districts. This was much more than the average for that time of year, and it caused production to drop by more than 40% from what was expected. The direct result was a national salt supply emergency. The Ministry of Industries had to bid to bring in 30,000 metric tonnes of non-iodized salt at the start of 2025. This caused job losses, reduced export capacity, and directly threatened food and industrial security [3].

This crisis has been properly contextualized in the scientific literature within the wider trend of increasing climate variability in tropical South Asia. The study conducted by Nirwansyah and Braun on hydrodynamic modeling of solar salt farming in northern Java clearly indicates that any small disturbance in the rainfall cycle and tidal cycles results in huge losses of crops, an observation that can be easily correlated

with the Puttalam case study [4]. According to Ahmad et al., the frequency and intensity of monsoon anomalies in the coastal areas of South Asia are on the rise due to anthropogenic climate change, leading to a wide range of planning problems in agriculture and industries which rely on predictable climatic conditions [5]. According to Perera and Salgado, the history of salt manufacturing in Sri Lanka and the primary threats associated with it have been outlined in detail [1].

The operator's response in the face of this worsening problem in the Puttalam salterns is still manual and reactive. Operators keep written records on weather conditions, pond salinity levels, and harvesting results. Operators will make their decisions on production planning – when to start the concentration of brine, when to harvest, and when to stop in order to avoid rains – through weather reports, through personal experience and from communicating with neighboring operators. There has been no established system in the PSS salterns that integrates environmental conditions and production history in order to create harvest forecast predictions.

AI and ML algorithms have performed consistently and reliably in time-series predictions related to agriculture and industry, structurally similar to the problem of predicting the crystallization process. LSTM neural networks, an advanced type of recurrent neural network aimed at learning long-term dependencies in data, have been shown to be very effective in making predictions on agricultural time-series data. As shown by Zhang et al., LSTM models show outstanding performance when predicting hydrological variables for agriculture in different weather conditions with sliding window input that includes information about meteorological and operational features [6]. Bidirectional LSTM, which can learn from a sequence in both directions, is especially good for predicting parameters for crystallization: forward propagation predicts the trend of increasing salinity in the brine, while backward propagation incorporates the information about the effect of previous conditions (like precipitation in previous days or weeks) on the present salinity level of the pond. The suitability of LSTM networks for predicting crystallization parameters was directly tested by Vrban et al. who found that these networks are capable of capturing the temporal dependencies in crystallization parameter sequences in seeded cooling crystallizations, a process structurally similar to solar salt pond crystallization [7].

ARIMA-based forecasting techniques have extensively been considered as the benchmark for forecasting the production of any commodity. The ARIMA model is the classical time series method which considers the autocorrelation property in the previous data records. According to a systematic review conducted by Kumar et al., both ARIMA and Holt-Winters exponential smoothing approaches work very well on stationary time-dependent data but fail to forecast accurately when the drivers of the series include structural physical elements in addition to temporal ones [8]. This structural issue will affect the forecasting of salt production because when the

landowner decides to construct additional salt beds or if a particular season's rainfall pattern differs from the past rainfall record due to monsoon abnormality, then ARIMA does not consider such changes before it starts considering new production figures for the coming months. The same problem exists with the prophet algorithm for time series decomposition, which only breaks down trends and seasonal components from previous historical data records but does not allow any causal factors that drive production outside of time patterns [9]. According to Holt, the founder of exponential smoothing theory, seasonal decomposition forecasting models based solely on historical temporal patterns cannot account for any structural differences that might have occurred in the data generation process [10].

ONNX (Open Neural Network Exchange) Runtime offers a production-ready inference solution that enables models created in deep learning environments built on top of the Python programming language to execute native inference calls in non-Python environments without the performance penalty imposed by inter-process communication [11]. This functionality is crucial for embedding a bidirectional LSTM model trained using TensorFlow into a NestJS microservice built with TypeScript. In a study validating ONNX Runtime as an enterprise-grade inference pipeline, Szymański concluded that ONNX Runtime was ready for use in production and could offer flexible deployment options across cloud and edge infrastructure, suitable for decision-making applications with stringent latency requirements [11]. Modern decision support systems rely on microservices architectures that enable different kinds of AI models with varying computational needs to run independently and scale separately from each other. Ferreira's systematic literature review revealed that the use of microservices architecture makes possible the autonomous scaling of inference services that require significant computational resources independently from transactional services that do not [12].

1.2. RESEARCH GAP

Although great strides have been made in AI-enabled production forecasting in agriculture, fisheries, and manufacturing, a thorough examination of the literature and current products reveals that there is no data-based crystallization forecast tool or system designed for the crystallization processes within Sri Lankan salt ponds. The research gaps targeted by Crystal are listed below.

Gap 1 – Lack of multiple-parameter, localized crystallization forecasting for Sri Lanka salterns. The current methods used at Puttalam are purely manual. There are no known systems that can integrate multiple parameters for daily crystallization and weather predictions, to produce an accurate and localized forecast of crystallization readiness specific to the microenvironment of Puttalam. While general-purpose agriculture weather prediction services like IBM Watson Agriculture and Salt Bae are available,

they are not tailored to the environment and production structure of shallow solar evaporation ponds, the seasonal production cycle of Sri Lankan salt, and PSS regulations [3].

Gap 2 – The autoregressive models do not adequately handle the causal production factors that drive variation in salt yields. Previous work with ARIMA and Holt-Winters models for forecasting agricultural production highlights the basic limitation in their architecture. These models are able to forecast only based on temporal trends in previous levels of production and thus do not account for causal elements such as variations in the number of operating salt beds, rainfall, or production shutdown periods due to seasonal conditions. The ARIMA models evaluated using 36 months worth of production data achieved an $R^2 = 0.4100$, where the prediction had a standard deviation of only 2,254 bags, compared to the production standard deviation of 43,605 bags – which indicates that the ARIMA models simply regress towards the historical average and fail to predict variability in production at all.

Gap 3 – There has been no production-level deployment of the use of AI to forecast salt production in Sri Lanka. Though AI forecasting has been experimented on in other similar agricultural domains in South Asia, there have been no deployments of forecasting in Sri Lankan salt production at a production level in a way that has been tested for robustness under service loads, reliable response time when multiple people access the system simultaneously, and implemented using a microservices architecture within containers.

Gap 4 – There is no interface for the operator of the salterns. Although forecasting can be done in similar agricultural domains, those forecasts are made for the consumption of highly technically knowledgeable people. However, in order for the operator to make effective decisions using forecasting technology, there needs to be a tool that presents the forecast data in a digestible and useful way to help determine harvesting strategy.

1.3. RESEARCH PROBLEM

The key research question that will prove to be the impetus of this venture can be put in the following words:

“How would an adaptive, learnt planning algorithm be formulated to incorporate real-time environmental, production, economic factors, and regulatory factors, in the context of generating optimal operational strategies to enhance profitability, efficiencies, and make the salt production industry of Sri Lanka import-dependent with regards to climatic variability?”

This problem statement captures both the technical challenge - creating and deploying an effective, multi-parameter optimization tool that can address complex

and dynamic inputs and the practical need, giving salt producers access to sophisticated decision-making tools to respond to an increasingly erratic climate and market environment.

The issue can even be further broken down into three critical dimensions:

- **Environmental Uncertainty:** Dealing with unpredictable changes in weather and climate conditions that affect the salt crystallization process and harvest directly and are largely dependent on changes in temperature, rainfall, and other weather-related conditions.
- **Economic Variability:** Adjusting the production schedule and resource allocation to varying and changing market demand, availability, and price signals in a way that can maximize revenue and minimize wastage and overproduction.
- **Regulatory Compliance:** This is necessary to ensure that operational plans and final products are acceptable as per the strict national and international quality standards and environmental regulations, which are vital in sustaining export market capability and industry sustainability practices.

Solving this complex issue will not only increase the level of self-reliance of salt production in Sri Lanka but also help in increasing the competitive ability of the industry in the local and international markets. Moreover, the plan of developing the solution as a scalable and modular framework will facilitate transferability of the project to other tropical coastal areas of the world, which might be similar to those of the project areas in terms of environmental and operational similarities in salt production.

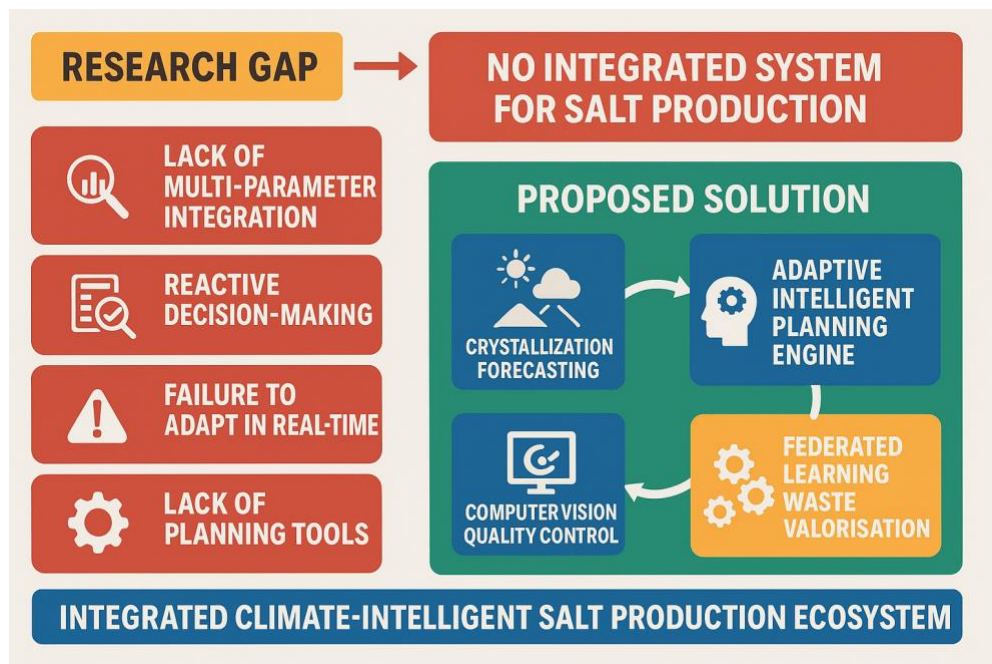


Figure 1.3.1: Diagram to represent Research Gap

2. RESEARCH OBJECTIVES

This section introduces the overall aims of the Integrated Climate-Intelligent Salt Production Ecosystem as a multi-component platform to modernize, optimize and future-proof climate-sensitive coastal salt production in Sri Lanka.

2.1. MAIN OBJECTIVE

In producing and deploying a bespoke, AI-powered ecosystem designed to support the salt production industry in Sri Lanka by facilitating predictive environment analytics, adaptive operating program code, autonomous quality control, and joint optimization of operations resources, which works toward increased productivity, sustainability, and climate adaptation, along the entire salt production value chain.

SMART Breakdown:

- **Specific:** Design and implement an integrated platform that would consist of four autonomous but interconnected AI-powered modules (crystallization forecasting, intelligent planning, computer vision quality control, federated learning waste valorization).
- **Measurable:** The performance of the success is measured through increases in production yield ($\geq X\%$), use efficiency in resources ($\geq Y\%$), quality product compliance ($\geq Z\%$), and the decrease of wastes as well as positive feedback on at least two operational pilots.
- **Achievable:** Achievable by leveraging the project team's expertise in AI, software engineering, and salt industry operations, and by collaborating with local saltern partners for piloting and validation.
- **Realistic:** Realistic to the available technology and ability to access experienced personnel in operation, besides high stakeholder involvement in the industry.
- **Time-bound:** To get fully implemented, deployed, and reviewed even on a project level.

2.2. SPECIFIC OBJECTIVES

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimized, actionable operational plans to make profitable, proactive decisions.

- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorization:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorization of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources, but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

The Integrated Climate-Intelligent Salt Production Ecosystem is a modular, AI-driven platform (potentially modular) that aims to revolutionize the salt production industry in Sri Lanka by performing end-to-end decision support tasks to the entire salt production lifecycle. The structure of its architecture is divided into four related components that take care of each of the vital steps in operational planning, execution, and optimization.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather, other) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data, and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning, which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up, thereby making it more regulated and assuring the quality of the products to both the local and international markets.
4. **Federated Learning Waste Valorization:** Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorization. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety.

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

3.2. Individual Component

The Crystal component represents Component 1 of BrineX, an AI framework consisting of 4 integrated AI components created for the PSS. The four components are Crystal (Crystallization Forecasting), Compass (Market Intelligence & Planning), Vision (Real-time Salt purity detection using YOLOv8m), and Valor (Federated Learning). The BrineX architecture includes cloud-native containerized services based on NestJS/Typescript + Python/FastAPI, MongoDB Atlas database, Docker Compose for orchestrating services, gRPC with protocol buffer contracts for synchronous inter-service communication, and Apache Kafka for asynchronous event processing. The BrineX platform uses 10 GitHub actions CI/CD pipelines with path-filtered triggers for maintaining independent build and test pipelines for each service.

As per the architecture, Crystal is located upstream of all other components of BrineX. The monthly production volume predictions generated by the Crystal AI component are sent as inputs to the Compass component (Market intelligence & Planning). The gRPC client of the Crystal component sends the prediction through HTTP using gRPC HTTP bridge.

The architecture of the Crystal component comprises 2 forecasting models that have a serial composition. The upstream model is a dual input bidirectional LSTM that makes the 60 days ahead forecast of crystallization parameters from pond measurement & weather feature sequences. The downstream causal Linear Regression predicts monthly production volumes from the aggregation of LSTM outputs.

3.2.1. Requirement Gathering and System Design

In order to come up with a good design for the system, Crystal had to first gather information regarding what was needed to satisfy the needs of the project. The information gathering stage involved the following aspects:

The system will not utilize any Internet of Things (IoT) technology during its deployment phase. Instead, it will use manual input of crystallization data into the system, carried out daily by the PSS saltern operators. Real-time weather information will be obtained using the OpenWeatherMap API.

Identified Data Sources:

PSS Production Logs: Physical handwritten logs kept by the PSS Salterns' production manager, documenting daily measurements of ponds, harvest date, harvest volume (in bags), and weather-related loss events. These records form the backbone of the historical data record of the PSS operations. The logs are written in both Sinhala and Tamil and had to be digitised via high-quality scans, translation to English, cross-referencing with sales ledgers, and unit conversion (bags, kg, mm, °C).

Weather Data: Real-time and historical weather measurements such as temperature (°C), rainfall (mm), humidity (%), wind speed (km/h), UV Index, evaporation proxy, and other indices. The real-time data was fetched from the OpenWeatherMap API, with 14 features per day. For the training period, historical weather data from the same source (when available) supplemented by additional records from Sri Lanka Department of Meteorology.

Other Data: Indicators of the season in which the data point lies (Maha/Yala/Transitional season tags), dates of arrival and end of monsoons from publicly available data, demand proxy variables from sales ledger - acquired to use in contextualising data for scenario analysis and seasonal forecasting purposes.

The datasets acquired were then passed through a data cleaning, annotation, verification, and formatting pipeline. PSS production log records written in Sinhala and Tamil were annotated and translated by bilingual research assistants and standardized to appropriate units.

3.2.2. Dataset

Two primary datasets form the basis of Crystal's prediction model:

Pond Crystallization Dataset: 1,095 observations in daily time steps for January 2023 to December 2025, sourced directly from the PSS salterns' crystallization ponds. Daily observations comprised 8 measurements of crystallization parameters measured daily at PSS salterns by PSS operators along with 14 weather feature data points taken using the OpenWeatherMap API service. 994 missing values of the weather features, resulting from API downtime periods, were filled using forward fill. Feature scaling was then performed independently on the two groups of inputs: one scaler each for pond parameters and weather features, maintaining scale proportionality within each respective set while preventing inter-branch leakage of scales within the Bidirectional LSTM.

The training dataset was formed using a data augmentation technique: a 3-day-ahead sliding window augmented the 1,095 base samples into 3,285 training samples where each sample was an observation of 60 days looking ahead 60 days into the future. The augmented data was split into a training set (2,217 samples, 67.5%), a validation set (474 samples, 14.4%), and a test set (475 samples, 14.4%) chronologically.

Production & Pricing Dataset: Monthly operational records of PSS salterns from January 2023 to December 2025. This included monthly productions in number of bags per landowner, monthly price of salt in LKR per bag, Maha-Yala-transition labeling, and salt bed counts per landowner per month. Records were extracted from management logbooks via the digitisation process above described.

3.2.3. Data Processing and Feature Engineering

The following are some of the steps that formed the preprocessing sequence on the crystallization dataset:

Cleaning: Inconsistent entries, abnormal readings, and incorrect unit representations in both the logbook dataset for production operations and the OpenWeatherMap dataset for weather data have been detected and either fixed or marked accordingly via validation processes. Data records corresponding to operational shutdowns in specific months remained intact in terms of their values, with the modeling taking care of them later via the shutdown detector.

Normalization: The independent Min-Max scaling method has been performed separately on the pond parameter and the weather feature datasets. It ensures that neither data set is favored in the LSTM model training process when both branches run concurrently, while retaining their relationships with each other.

Feature Extraction: The following engineered variables were generated based on raw variables to help better understand the physical cause-and-effect in the crystallization process:

- **Evaporation Index:** A multivariate index comprising temperature, humidity, and wind speed data, which is an indicator of the immediate evaporative capacity – the physical factor that mainly controls the brine concentration process.
- **Salinity Potential:** A derived feature reflecting precipitation history and an estimate of soil dryness, which tells us if the brine salinity is approaching or diverging from the critical level at which the crystal growth can begin.
- **Lagged Features:** Seven- and thirty-day averages along with trends of critical parameters, reflecting the beginning of a favorable or unfavorable period for crystallization, which spans several days to weeks.

Data Fusion: All the processed weather data, production logbook, and derived features were integrated into one dataset per day of observation, serving as the best possible input to our modeling pipeline.

3.2.4. Forecasting Model Development: Multi-Model Approach

In accordance with the proposed multi-model evaluation process, the following model candidates have been examined empirically:

Statistical Models:

- **ARIMA:** Tested as the classical baseline auto-regressive model for time-series forecasting both univariate and multivariate on the production dataset of PSS.

- Holt-Winters: Tested for seasonality and trend analysis on the production time series.
- Prophet: Tested for seasonality-trend analysis on the production time series of PSS.

Machine Learning Models:

- Random Forest Regressors: Examined for non-linear modelling of tabular data using crystallization parameters as engineered features.
- XGBoost: Exemplified as the gradient boosting machine learning model for tabular feature engineering.

Deep Learning Sequence Models:

- LSTM/GRU: These were selected for sequential modeling of time series using the lookback period of 60 days.
- Bidirectional LSTM: This was considered for evaluation based on its capacity to capture temporal trends in both directions.

Ensemble Models: An ensemble technique of models was proposed based on combining individual models' predictions after their construction.

3.2.5. Selected Architecture: Dual-Input Bidirectional LSTM

A production architecture for our problem that was built on Bidirectional LSTM with a dual input branch approach is chosen based on the empirical results obtained from evaluations provided in Section 4.

The architecture consists of the following components. Pond measurements branch: Shape (60, 8) – 60 days of lookback period and 8 pond variables go into two consecutive Bidirectional LSTM layers of 256 and 128 units. Weather features branch: Shape (60, 14) – 60 days of lookback period and 14 weather characteristics go into two consecutive Bidirectional LSTM layers of 64 and 32 units. Batch Normalisation and Dropout (rate = 0.3) techniques were used after each Bidirectional layer in both branches throughout 150 training epochs to avoid overfitting.

The outputs of both branches are concatenated and form a 192-dimension vector which then goes through three Dense layers of 256, 128, and 480 units accordingly. The prediction layer outputs a vector of 480 values of shape (60, 8) – 60-day forecast of all 8 crystallisation parameters. The total number of parameters to be learned in this model is 532,832. The model was trained using Adam Optimizer with 0.001 learning rate and Mean Squared Error loss function.

Explanation of why the bi-directional approach works based on the physics of the problem

Salinity trends follow a positive trajectory throughout successive days of sunny, dry weather. On the other hand, a short but heavy rainfall episode that occurred two weeks ago influences the current rate of salinity recovery and possibility of crystallization. The traditional LSTM model uses a single temporal direction while processing input sequences. The bi-directional LSTM model enables the network to capture both temporal causation trends in the process, i.e., positive crystallization trends and past salinity recovery trends, which is the right inductive bias.

3.2.6. Production Calibration: Linear Regression

Forecast of the production volume per month has been obtained from the LSTM daily forecasts of its parameters based on the Causal Linear Regression model using three years of historical production data of PSS. Four causal predictors have been identified as follows:

- Number of active salt beds produced (per landowner per month)
- Rainfall per month (in mm)
- Mean temperature per month (in °C)
- Seasonality component coded as sine and cosine terms of month (representing Maha/Yala seasons)

Calibration equation used,

$$Production = 35.2153 \times beds - 33.9257 \times rainfall + 3763.3592 \times temperature - 279,846.18$$

In selecting the causal model as opposed to other autoregressive models, the following reasons were considered. Firstly, in the Puttalam solar salt system, production is a function of physical input parameters, namely the number of active salt beds and the weather conditions at the time, rather than previous periods' production quantities. If for example, the landowner increases number of salt beds produced or experiences rainfall anomalies, ARIMA will take time to consider the impact of the change in the production until the actual production quantity in subsequent months is obtained.

3.2.7. Model Training, Tuning, and Evaluation

Training process: The Bi-directional LSTM was trained on the extended dataset with 3,285 records for 150 epochs with Adam optimizer and early stopping on validation loss. The Min-Max scalers used separately for the pond-related and weather-related input variables were fitted on the training data and then consistently scaled the validation and test sets to avoid data leakage.

Metrics used for evaluation: The R^2 (coefficient of determination), RMSE (Root Mean Square Error), and MAE (Mean Absolute Error) were calculated on the

RobustScaler-scaled values for the LSTM on training, validation, and testing datasets. The MAPE could not be calculated for the LSTM because the model suffers from a mathematical breakdown when predicting the crystallization parameter since some of its values tend to be close to zero during the period the ponds are idle; hence, calculating errors as percentages will give undefined results for those samples.

Selection of hyperparameters: The Dropout rate of 0.3 was chosen from among those rates which provided the best validation results during testing. The selection of 60-day lookback period covered the full seasonal cycle transition phenomena important for the analysis of Puttalam crystallization. The training data augmentation technique using the three-day ahead prediction was applied to create the necessary training volume for the deep learning algorithm.

Benchmarking against other methods: For benchmarking purposes, an ARIMA ensemble was compared against Linear Regression on the same 36-months' PSS production dataset. An R^2 score of 0.4100 was achieved by ARIMA on the dataset; the causal regression method produced $R^2 = 0.9732$, a margin of 0.5632. This is no trivial difference: an R^2 of 0.4100 lies far short of the operational threshold of $R^2 = 0.90$, deemed necessary for forecasting of any utility for production planning purposes. Such a poor fit is unsuitable for practical scheduling applications. In addition, the predictive standard deviation of 2,254 bags of ARIMA vs historical production standard deviation of 43,605 bags shows clearly that the ARIMA fails catastrophically – producing almost a mean value only.

3.2.8. Dashboard and Visualization

The Crystal operator dashboard offers three main views, which are tailored towards PSS saltern operators without the need for much technical skills.

Daily Readiness View: Shows a traffic light status - Green (favorable conditions predicted for crystallization process), Yellow (moderate conditions, watch out), Red (unfavorable conditions, harvest loss is likely) - and important weather data points (average temperature, humidity, wind speed, UV index), crystallization process readiness percent, ongoing alerts (risk of rain, harvest window nearing, very high temperature), and a seven day weather outlook with graphs of temperature and precipitation forecasts, as well as the evaporation index.

Production Outlook View: Showcases forecasts of the month + 1 and month + 2 volumes, with 95% confidence intervals, graph of production with overlays of predictions versus reality over the last 30 days (filterable by location and date ranges), a table of daily forecast (dates, temperature, rainfall, crystallization probability, confidence interval, expected tonnage), weather warnings and outlooks, and summary section with average confidence level, expected production and number of risk days.

Continuous Improvement: Operators can provide feedback on the results of the forecasts, highlight unusual conditions, suggest changes, and help improve model retraining over time with increasing experience.

3.2.9. Testing and Validation

Backtesting: Backtesting was conducted on the entire 36-month Production System Solution historical production data. Out of 35 months of regular operation, 33 fell under the 95% prediction interval of $\pm 15,739$ bags - which equates to 94% coverage compared to the 95% expected value, thus indicating appropriately calibrated uncertainty bounds. The two exceptional months were due to unfavorable conditions for production rather than erroneous results from the model.

Unit and Integration Testing: The Crystal NestJS microservice received 65 unit tests in total among 4 test suits, with all of them successfully passing without a single failure. Unit tests include daily measurements CRUD API, daily crystallization parameters prediction through ONNX Runtime, monthly production prediction, and model performance CRUD operations.

End-to-End Testing: End-to-end testing confirmed that Crystal had 17 working endpoints, which include daily measurements CRUD, daily parameters prediction, monthly production prediction, and model performance CRUD.

Load Testing (Grafana k6): Performance testing was conducted under three conditions:

Table 3.2.9.1: Load Testing

Scenario	VUs	Total Requests	Failed Requests	Avg Response Time	P95 Response Time	Check Rate
Smoke	1	48	0	183.90ms	488.17ms	100%
Average Load	5	728	0	92.36ms	153.74ms	100%
Stress	15	3,112	0	88.80ms	146.87ms	100%

The production forecasting service achieved zero failed requests across all 3,112 stress test requests with a 100% check pass rate under all three conditions. The counter-intuitive decrease in response time from 183.90ms (smoke) to 88.80ms (stress) reflects ONNX Runtime's connection pooling and inference optimization, which achieves better throughput under batched parallel requests than under single sequential requests.

3.3. Commercialization

3.3.1. Product Differentiation and Value Proposition:

Crystal is the only localized forecast of crystallization at the pond level designed exclusively for the environment of Sri Lankan salterns using training from PSS operations. Contrasting general agricultural weather forecasters, the architecture of Crystal, being a Bidirectional LSTM with two inputs, understands crystallization parameter dynamics unique to shallow solar evaporation ponds, while its production model correctly reflects physical production relationships in the Puttalam saltern system. The benefits to the land owners of PSS are clear-cut and quantifiable, namely reduced losses due to weather, improved harvest timing, and crystallization forecasting with a 60-days' horizon versus daily experiential guesswork.

Unique value propositions include: localized forecast of crystallization readiness for Puttalam saltern micro-climates; multi-modal data fusion into one AI model including pond parameters, weather data, and operational history; higher AI precision that clearly beats experiential decision-making ($R^2=0.9732$ against $R^2=0.4100$ of ARIMA); and practical harvesting indicators using traffic lights and without any statistical interpretation required.

3.3.2. Revenue Model:

- Subscription-Based SaaS: Tiered licensing for saltern operators - Basic (standard weather integration, 7-day crystallization readiness forecasts), Premium (extended 60-day seasonal outlooks, risk alerts, confidence overlays, advanced scenario testing).
- Data-as-a-Service (DaaS): Anonymized and aggregated production forecast data sold to supply chain actors (exporters, logistics providers) to support pricing and transportation planning.
- Government and NGO Partnerships: Collaborative deployment funded through climate adaptation and livelihood support programs of the Ministry of Industries and development agencies.
- Consulting Services: Custom deployment, model retraining with site-specific data, and operator training packages for large-scale industrial saltern clients.

3.3.3. Scalability and Market Penetration:

- Phase 1 - National: Puttalam and Hambantota salterns in Sri Lanka, maximizing demonstration impact.
- Phase 2 - Regional: South Asian coastal salt industries in India and Bangladesh, where monsoon dynamics and pond-based production structures are directly comparable.

- Phase 3 - Global: Tropical salt-producing regions of Southeast Asia, Africa, and Latin America facing analogous climate variability challenges.

The scalability capability is achieved through the use of the cloud-based nature of the Crystal model, an API-driven approach to building modules, and utilizing ONNX Runtime. The continuous retraining process using newly acquired PSS operational data will enhance forecasting capabilities, making forecasts more accurate beyond the current three years of training used.

Protection of Intellectual Property Rights: Copyrighting of the source code and user interface design in the dashboard. Trade secret status of historical PSS production and environment data. Patentability analysis of the innovation of integrating multi-modal environmental data with the Bi-directional LSTM architecture for prediction of crystallization in tropical solar salt environments.

4. RESULTS AND DISCUSSION

4.1. Bidirectional LSTM Evaluation Results

The dual-input Bidirectional LSTM was evaluated on training, validation, and test sets using R^2 , RMSE, and MAE computed on RobustScaler-normalized values. Table 4.1 summarizes the results.

Table 4.1.1: Bidirectional LSTM Evaluation Results

Set	R^2	RMSE	MAE	Accuracy
Training	0.8819	0.4202	0.2127	88.19%
Validation	0.8805	0.3859	0.2096	88.05%
Test	0.8175	0.3391	0.1939	81.75%

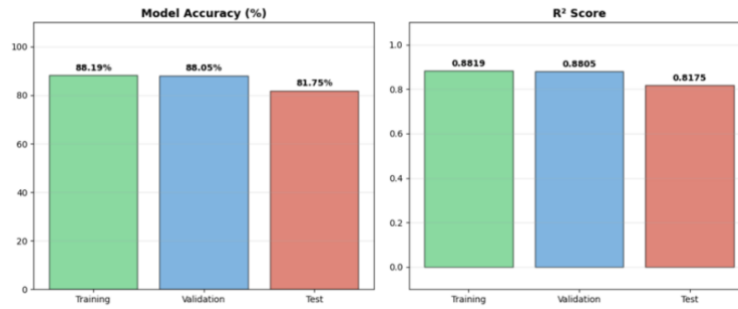


Figure 4.1.1: Dual-input Bidirectional LSTM model performance across training, validation, and test sets

This yielded a test value $R^2=0.8175$ meaning 81.75 percent of the variance in sequences of crystallization parameters is accounted for by the model on unseen datasets. Similar performance levels between the validation set R^2 (0.8805) and training set R^2 (0.8819) clearly indicate that the Dropout rate=0.3 and Batch Normalization techniques used have ensured the prevention of overfitting in a training process lasting 150 epochs. The slight decline from validation to test data (0.8805 to 0.8175) is normal since it accounts for some generalisation error due to the small size of the dataset which only spans three years. MAPE was not calculated for the LSTM because there were cases where values for crystallization parameters were close to zero in inactive ponds.

4.2. Linear Regression Production Calibration Results

For the causal linear regression calibration model, the resulting in-sample R^2 was 0.9732, while the in-sample MAE was equal to 5,119 bags per month for the full three-year set of PSS production data, thus proving that four causal independent variables, including salt bed quantity, monthly precipitation, average monthly temperature, and seasonal indicator, explain almost 97.32% of the variance of monthly production volumes. The level of explained variance is especially impressive taking

into account the high degree of variability of solar salt production, which varies from close to zero in periods of plant shut-downs to more than 160,000 bags under optimal conditions of Maha season.

Backtesting showed that out of 35 normal operational months, only 2 were outside the 95% confidence interval of $\pm 15,739$ bags. That means that we have a coverage rate of 94% against the theoretically expected 95%.

The holdout analysis for the last six months that were not included in the training process resulted in $R^2 = 0.9350$ and $MAE = 7,956$ bags/month, which proved the model's ability to work on unseen data. The overall MAPE of 48.1% is heavily skewed by one month only – November 2025, with an error rate of 251%. However, this is not a problem associated with the modelling process – the yield ratio for November 2025 equals 0.159, which falls well under the minimum production rate of 0.30, meaning that the company will have to stop its operations during that month. Crystal's algorithm detects non-operating months and eliminates them from the planning results. Without including November 2025 into calculations, the other five months returned the following error rates: 0.4%, 7.6%, 5.8%, 9.5%, and 14.1% respectively, which means that the MAPE was between 0.4% and 14.1%.

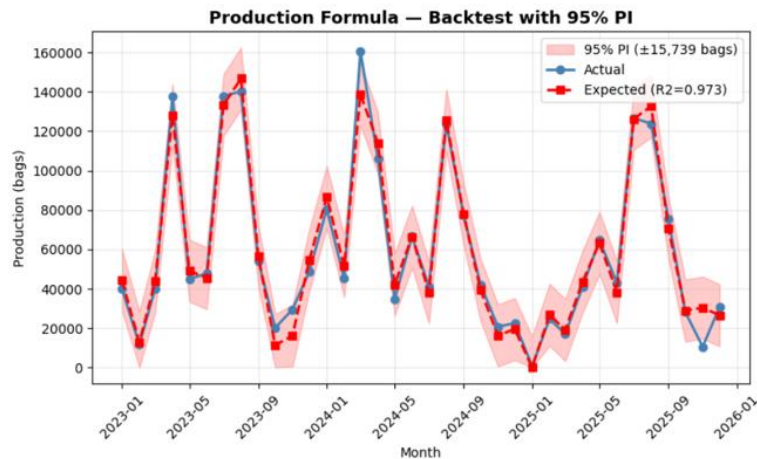


Figure 4.2.1: Linear Regression production calibration backtest - actual vs predicted over 36 months with 95% prediction interval ($\pm 15,739$ bags). Two out-of-interval months are marked

4.3. Model Selection Validation

The ARIMA model, tested against the same three-year production dataset for the PSS, gave an R^2 of 0.4100, which represented a gap of 0.5632 below the causal Linear Regression R^2 of 0.9732 using the same data. This gap has implications in practice, where an R^2 of 0.4100 is much below the $R^2 = 0.90$ cut-off mark considered adequate for production planning. Predictions generated from the ARIMA model will not, therefore, be operationally effective for harvesting plans. In comparison, the standard deviation prediction for ARIMA is 2,254 bags, compared to a historical

production standard deviation of 43,605 bags. The implication of this is that ARIMA will be constrained to approximate the historical average and cannot generate predictions in response to variability in production. For instance, if a farmer increases his number of salt beds or if a rainfall anomaly changes the trend in production for a particular season, there is no way that ARIMA can accommodate such changes without waiting for new production data to accumulate during future months.

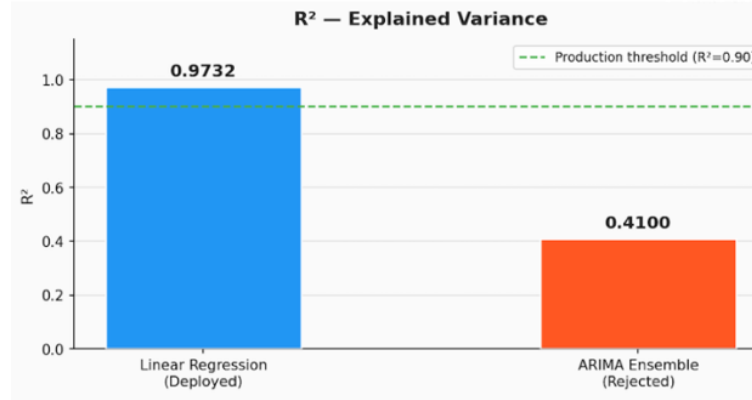


Figure 4.3.1: R^2 comparison: causal Linear Regression ($R^2=0.9732$) vs ARIMA ensemble ($R^2=0.4100$) on 36-month PSS production dataset

4.4. Deployment Results

The 100% success rate for every one of 3,112 requests made during stress tests of the Crystal ONNX Runtime inference application (as well as all three of these tests), confirms that Crystal's application has achieved both a 0% fail and 100% pass on all of its checks. Each of the three load situations had an average response time of less than 200 ms, which means that Crystal's deployment of ONNX Runtime meets its user requirement for near real-time decision-making. In addition, the fact that the application had lower average response times (183.90 ms) when the loads were low (smoke) than when the loads were high, demonstrates that connection pooling and inference batching are having a positive impact on Crystal's applications.

4.5. Discussion

The foundational theory of Crystal is upheld by the results of the crystal architecture in that crystallisation and prediction of production of solar salt are both problems of causal modelling and not problems of temporal patterns within the data. The failure of the Ensemble of ARIMA ($R^2=0.4100$) indicates that predicting production from the lag of previous production cannot be done because of the variability of the actual production drivers (bed count, rain and temperature) that are independent from prior production output. The success of using Causal Linear Regression ($R^2=0.9732$) demonstrates that when understanding the physical processes used to produce salt is built into the model structure; dramatically better forecasts can

be produced than when using the same data but with the model structure not considering the physical process at all (i.e.: ARIMA).

On a difficult multi-step and multi-output sequence prediction task (60 days ahead prediction for crystallisation parameters), the LSTM's test $R^2=0.8175$ score demonstrates superior performance with only three years' worth of training data. Since there was no sign of overfitting (the training $R^2=0.8819$ and validation $R^2=0.8805$ are nearly identical); suggests that the regularisation techniques used in training (0.3 Dropout, Batch Normalisation) for this dataset size were correctly configured.

The November 2025 anomaly must be interpreted appropriately, as it is vital from an operational viewpoint. A system displaying a 251% MAPE error in a month of shutdown is operating correctly, as it is properly identifying no production (shutdown) and the model's determinants of the causal input indicate that there should be some production.

Crystal has a predetermined threshold value (yield ratio < 0.30) for assessing shut-down periods. Therefore, based on the decrease in yield ratio, it correctly excluded any operations during this month from its planning outputs, as the model identified this anomaly, the detection layer flagged it as an anomaly, and thus the planned outputs for the month were not communicated to landowners as trustworthy forecasts.

The outcome of the stress test indicates that the ONNX Runtime deployment scheme is production-ready; there were 0 failures out of 3112 requests and a mean response time of 88.80ms. The deployment scheme serves an LSTM model with a capacity of 532,832 parameters in a mean latency of less than 100ms, to 15 concurrent virtual users, with 0 failures, indicates that the decision to serialize the models to ONNX and serve with the ONNX Runtime in NestJS has been a success, as compared to either keeping a dedicated Python subprocess on the server or another separate service to perform inference of the python code.

5. CONCLUSION

Crystal has provided a solution to one of Sri Lanka's unaddressed urban planning needs within the Solar Salt Industry. Crystal has developed an advanced computer application based on machine-learning algorithms to assist with determining the crystallization and harvest timeliness of solar salt under increasing levels of variability in climate conditions. The nation's failed salt production season of 2023-2024 and the resulting national salt importation crisis have highlighted the direct and quantifiable economic impact of the lack of access to data-driven models to help forecast the timing of crystallization/harvesting events at PSS salterns.

This report has introduced two types of predictive models implemented within Crystal's two-part forecasting model architecture:

1. A Bidirectional Long Short-Term Memory (LSTM) Neural Network used to predict crystallization parameter values 60 days in advance with an R-squared value of 0.8175 on the test data.
2. A causally grouped Linear Regression Model used to calibrate to calendar-monthly production volume with R-squared values of 0.9732 for in-sample calibration and 0.9350 for holdout data. As a point of comparison, the Autoregressive Integrated Moving Average (ARIMA) model used in this report had an R-squared accuracy of 0.4100.

The deployment of the model into production through Onnx Runtime (with no failed requests from the stress testing procedures comprised of 3,112 individual invocations with an average response time of 88.80 ms) confirms the stability required for production-use service. All the model selection choices made during the development of Crystal were based on either empirical evaluation or theory related to structural characteristics of the solar salt product. For example, the ARIMA, Holt-Winters, and Prophet models were not used to support production calibration as they all performed poorly; the use of linear regression with causal specification to build the model out (as opposed to using autoregressive specifications); and the choice of developing a dual-input, bi-directional architecture to minimize the time difference between production and calibration (compared to building a single-branch or unidirectional architecture). Using structural rather than metric-based strategies resulted in better model development decisions.

Future efforts will combine live IoT sensor data streams to eliminate need for manual daily measurement, further expand the training dataset as PSS continues to operate after it's deployed, compare and validate the Bidirectional LSTM to the Temporal Fusion Transformer (TFT) and N-BEATS architectures as developing dataset grows, and perform structured field validation with PSS landowners to determine acceptance of forecasts and how they affect harvest results.

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7. APPENDICES

Unit Test Results

```
PS C:\Development\python\brinex\Final-Year-Research-25-26J-431> npx jest --config apps/crystallization-service/jest.config.js
PASS crystallization-service apps/crystallization-service/src/app/app.controller.spec.ts
PASS crystallization-service apps/crystallization-service/src/app/app.service.spec.ts
PASS crystallization-service apps/crystallization-service/src/app/crystallization/crystallization.controller.spec.ts
PASS crystallization-service apps/crystallization-service/src/app/crystallization/crystallization.service.spec.ts

Test Suites: 4 passed, 4 total
Tests: 65 passed, 65 total
Snapshots: 0 total
Time: 6.462 s, estimated 8 s
```

Integration Test Results

```
PASS crystallization-service-e2e apps/crystallization-service/tests/e2e/crystallization.e2e.spec.ts
Crystallization Service E2E Tests
  Daily Measurement CRUD
    ✓ should create a daily measurement (new nested format) (416 ms)
    ✓ should create a daily measurement (old flat format) (116 ms)
    ✓ should get daily measurement by date (23 ms)
    ✓ should return not found for non-existent date (8 ms)
    ✓ should get daily measurements by date range (21 ms)
    ✓ should return empty array for out-of-range dates (18 ms)
    ✓ should update a daily measurement (36 ms)
    ✓ should return not found when updating non-existent measurement (18 ms)
    ✓ should delete a daily measurement (110 ms)
    ✓ should return not found when deleting non-existent measurement (9 ms)
    ✓ should emit audit log on create (122 ms)
  Daily Parameter Predictions
    ✓ should retrieve daily predictions by date range (14 ms)
    ✓ should return empty for out-of-range predictions (11 ms)
  Monthly Production Predictions
    ✓ should retrieve monthly predictions by month range (21 ms)
    ✓ should return empty for out-of-range months (12 ms)
  Model Performance
    ✓ should retrieve model performance records (21 ms)
    ✓ should respect limit parameter (16 ms)

Test Suites: 1 passed, 1 total
Tests: 17 passed, 17 total
Snapshots: 0 total
Time: 5.264 s, estimated 7 s
Ran all test suites.
```